NHS
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Integrated Care Board

Adult Bronchiectasis Primary Care Clinical Pathway

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NCL-Wide / Borough(s)	NCL-Wide
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Key

Pi	**"Must do" actions for GP's / (Triaged by RSS)**
B	**Recommendations for Primary Care**
R	**Red flag / urgent referral**
O	**Routine referral**
P	**Public health intervention**
G	**Audio-visual aids for patients and GP**
📄	**Click icon for clinical evidence**
○	**Right-click to use hyperlink**

Note: Alpha-numeric step references are to aid printing in black & white and colour blindness

Investigations must be within 6 months of date of referral

1.0 (B)

Patient presents with following symptoms suggestive of bronchiectasis:
- Chronic cough with daily mucopurulent sputum
- Frequent lower respiratory tract infections (LRTI) especially if immunocompromised
- Breathlessness
- Co-existing chronic rhinosinusitis
- Haemoptysis: typically mild with bloody specks in the sputum
- Asthma that is not responding to treatment
- COPD with 2+ exacerbations/year & previous pseudomonas aeruginosa
- Rheumatoid arthritis if they have symptoms of chronic productive cough or recurrent chest infections.
- Inflammatory bowel disease and chronic productive cough.

2.0 (B)

Take full respiratory history including presenting symptoms:
- Nature of cough and associated upper airway features, e.g. sinusitis, middle ear infection
- Characterise sputum; volume, thickness and colour and whether haemoptysis
- Asthmatic features, shortness of breath, wheezing, chest pain
- COPD, recurrent LRTI and previous pneumonia
- History of smoking tobacco and other substances (note secondary tobacco smoking exposure)

Past medical & family history:
- Age of onset and duration of symptoms
- General systemic health including, weight loss, gastrointestinal symptoms, particularly symptoms of gastroesophageal reflux disease, rheumatological symptoms and/or subfertility
- Other infections e.g. conjunctivitis, UTI, skin or soft tissue infections

3.0 (Pi)

Investigations:
- Chest X-ray to exclude other pathology and to help confirm the diagnosis where disease is severe.
- Sputum microbiology
- Spirometry to assess the severity of airflow obstruction and identify any co-existent diagnoses
- FBC and oxygen saturation levels

4.0 (B)

Exclude differential diagnoses

5.0 (B)

- Suspected asthma: See asthma pathway (coming soon)
- Suspected COPD: See COPD pathway (coming soon)

6.0 (O)

If bronchiectasis suspected clinically, please refer for specialist respiratory outpatient assessment to:
- Confirm diagnosis with a high resolution CT scan (50% do not have any CXR signs suggestive of bronchiectasis)
- Investigate cause e.g. Allergic Bronchopulmonary Aspergillosis, Non-tuberculous mycobacteria (NTM), tuberculosis, hypogammaglobulinaemia, cystic fibrosis, and primary ciliary dyskinesia
- Agree management plan
- Start teaching and enabling effective self-management

7.0 (R)

Red flag symptoms 2WW:
- Unexplained haemoptysis (age ≥ 40 years)
- Unexplained weight loss
- Unexplained appetite loss
- Deep vein thrombosis
- Blood stained mucus
- Prominent systemic features (myalgia, fever, weight loss)
- Unexpected clinical findings (eg crackles, clubbing, cyanosis, cardiac disease, monophonic wheeze or stridor)
- Chest X-ray shadowing
- Persistent non-variable breathlessness

8.0 (G)

Lifestyle advice popup

Smoking status = smoker - (Ask Advise Act)
Alcohol consumption = <14 units/per week
Physical inactivity / diet, aim for BMI <30, <150 mins per week
Mental wellbeing = Signs of stress / anxiety
Refer to GP website for further information on local services

8.1 (P)

9.0 (G)

PATIENT INFORMATION

Patient information can be found at: http://www.blf.org.uk/support-for-you/bronchiectasis [1]

9.1 (P)

10.0 (B)

Review and management of patients with bronchiectasis in primary care
Review at least annually. Frequency should be determined by action plan
Review should include:
- Smoking status including checking for other smoked drugs e.g. cannabis
- Check number of exacerbations in the last year. Refer to specialist if exacerbations are causing significant morbidity, hospitalisation or three or more infective exacerbations a year
- Assessment of symptoms and how they affect daily living
- Review of sputum volume, character and undertake sputum culture and sensitivity test to assess for chronic bacterial colonization
- Offer and refer for pulmonary rehabilitation if symptomatically limited by breathlessness (MRC Breathlessness score =2; short of breath hurrying on the same level) or if patient is regularly productive of sputum to teach chest clearance techniques. Assess compliance with sputum clearance exercises
- Spirometry and oxygen saturation
- Measurement of BMI — if the person's BMI is less than 20 seek specialist nutritional advice.
- Treat any co-morbid conditions e.g. asthma/COPD, rhinitis/sinusitis and acid reflux
- Offer annual influenza vaccination (including to household contacts) and ensure up to date with COVID vaccinations and pneumococcal vaccination (Pneumovax® 23). Please note revaccination with pneumococcal vaccination is not necessary unless person is asplenic or has chronic renal disease
- **For management of exacerbations please refer to managing exacerbations of bronchiectasis pathway (see below)**

11.0 (O)

Refer to specialist team if:
- 3 or more exacerbations per year or hospitalisation
- Detection of pseudomonas, other resistant organisms or allergic bronchopulmonary aspergillosis
- Decline in FEV1
- Long term antibiotic therapy
- Associated rheumatoid arthritis, immune deficiency, inflammatory bowel disease, or primary ciliary dyskinesia.
- Advanced disease and those considering lung transplantation

See bronchiectasis severity index, detailed in British Thoracic Society guidelines (https://www.brit-thoracic.org.uk/quality-improvement/guidelines/bronchiectasis-in-adults/)

Managing exacerbation of bronchiectasis in Primary Care

12.0 (B)

Exacerbations
(who needs antibiotics?)

13.0 (B)

No antibiotics
Mucoid or purulent sputum but patient stable and well

14.0 (B)

Antibiotics (NICE Bronchiectasis (non-CF) antimicrobial guidance) [2]
Purulent sputum with increased volume, thickness or 'darker' blood stained sputum
Feels less well: worsening breathlessness, new fever or chest pain, malaise, drop in oxygen saturation or spirometry readings.

15.0 (R)

Arrange hospital admission for adults who:
- have signs of cardiorespiratory failure or sepsis
- are cyanosed or acutely confused
- marked breathlessness, rapid respiration, or laboured breathing
- peripheral oedema
- temperature of 38°C or higher

16.0 (Pi)

Always send a sputum first and specify bronchiectasis on the form and request culture for pseudomonas aeruginosa, BUT don't wait for result before starting antibiotics.

17.0 (B)

Oral antibiotics - Patient should have a reserve course (based on last sputum culture) at home for use when needed
Ensure patients and carers who are able to manage medication have been advised to start antibiotics immediately on exacerbation. They should:
- Understand when it is appropriate to start treatment
- The importance of collecting sputum before starting treatment and seeking medical advice

18.0 (B)

Reassess people with an acute exacerbation of bronchiectasis if their symptoms worsen rapidly or significantly at any time

Note: usual treatment course of antibiotics is 7-14 days

No previous *Pseudomonas aeruginosa*
- Amoxicillin 500mg tds for 7-14 days OR
- Doxycycline 200mg stat then 100 mg od for 6-13 days OR
- Clarithromycin 500mg bd 7-14 days OR

Consider alternative choice antibiotic e.g. co-amoxiclav 625 mg tds for 14 days if higher risk of treatment failure. People who may be at higher risk of treatment failure include people who've had repeated courses of antibiotics, a previous sputum culture with resistant or atypical bacteria, or a higher risk of developing complications
First isolation of *Pseudomonas aeruginosa*
Refer for specialist respiratory review
Ciprofloxacin 750mg bd for 7-14 days*
Previous *Pseudomonas aeruginosa*
Ciprofloxacin 750mg bd for 7-14 days*
(do not prescribe second course within 12 month period unless advice and guidance is sought see below)
Staphylococcus aureus
Flucloxacillin 500mg qds 7-14 days OR
Clarithromycin 500mg bd 7-14 days
Non-tuberculous mycobacteria (NTM)
If found REFER *for specialist respiratory review*

*Safe prescribing of fluoroquinolones (See NCL position statement for full alert)[3]:
- Do not use fluoroquinolones in patients with known aortic aneurysm, aortic dissection, congenital heart valve disease, pre-existing heart valve disease or history of serious side effects related to quinolone treatment (including tendon disease/disorder) unless there are no other treatment options available
- Minimise course length in high risk group (7 days and reassess). See NCL position statement [3] for details of high risk groups. Provide patients prescribed fluoroquinolones with information about risks associated with fluoroquinolones. Consider providing MHRA patient information on the side effects of tendons, muscles, joints & nerves [4]
- Counsel on discontinuation criteria (as per NCL position statement) [5]

18.1 (Pi)

19.0 (R)

If not appropriate to prescribe ciprofloxacin to treat pseudomonas, or not improving/still unwell (including fatigue and weight loss) after appropriate course of oral antibiotics, even if sputum culture negative or normal inflammatory markers refer to:
- Ambulatory care for consideration of intravenous (iv) antibiotics including option of home iv treatment, and
- Specialist respiratory outpatient assessment.

20.0 (B)

Follow-up of infective exacerbation. Review the following:
- response to empirical treatment when sputum culture and sensitivity results are available:
  - if the patient is responding well, continue with the prescribed antibiotic – do not change the treatment based on culture results
  - if the patient has not responded well to treatment, prescribe a different antibiotic
- effectiveness and acceptability of the chosen airway clearance technique within approximately 3 months of the initial exacerbation

If the patient deteriorates at any stage after starting treatment, reassess to see if hospital admission is indicated

21.0 (O)

Refer to specialist adults who:
- Have symptoms that are not improving with repeated courses of oral antibiotics.
- Have bacteria in their sputum that are resistant to the available oral antibiotics.
- Are unable to tolerate oral therapy.

Refer the following patients to specialist:
- deteriorating bronchiectasis with worsening spirometry (FEV1)
- colonised with chronic Pseudomonas aeruginosa, opportunist mycobacteria or methicillin-resistant Staphylococcus aureus (MRSA)
- recurrent LRTI - > 3 per annum
- bronchiectasis and associated rheumatoid arthritis, immune deficiency, inflammatory bowel disease, and primary ciliary dyskinesia (PCD)
- allergic bronchopulmonary aspergillosis (ABPA)

22.0 (B)

Annual review
(see pathway above)

23.0 (G)

PROPHYLACTIC ANTIBIOTICS

Consider Advise or Guidance or referal for specialist respiratory outpatient assessment for consideration of prophylactic antibiotics if poorly controlled/progressive disease ie:
- needing ≥ 3 antibiotic courses in one year
- needing a course of iv antibiotics
- rapid relapse without an explanation

24.0

[1] **Patient information**
http://www.blf.org.uk/support-for-you/bronchiectasis
[2] **(NICE Bronchiectasis (non-CF) antimicrobial guidance**
https://www.nice.org.uk/guidance/ng117
[3] **NCL Position statement**
https://www.ncl-mon.nhs.uk/wp-content/uploads/Guidelines/5_Fluoroquinolones_Safe_Prescribing_Position_Statement.pdf
[4] **Patient information on the side effects of tendons, muscles, joints & nerves**
https://assets.publishing.service.gov.uk/media/5c9364c6e5274a48edb9a9fa/FQ-patient-sheet-final.pdf
[5] **NCL Position statement**
https://www.ncl-mon.nhs.uk/wp-content/uploads/Guidelines/5_Fluoroquinolones_Safe_Prescribing_Position_Statement.pdf